

Genomic Cancer Report

Comprehensive somatic mutation analysis · Solid tumor panel v4.1

PATIENT

Jane Smith, 54F

SAMPLE ID

TUM-2024-04821

TUMOR TYPE

Non-small cell lung cancer

REPORT DATE

2026-05-28

ORDERING PHYSICIAN

Dr. K. Tanaka

TUMOR PURITY

72%

Summary

VARIANTS DETECTED

4

3 pathogenic · 1 VUS

ACTIONABLE VARIANTS

3

FDA-approved therapies available

TMB

12.4 mut/Mb

Intermediate

MSI STATUS

MSS

Microsatellite stable

Detected variants

COSMIC

ClinVar

OncoKB/CIVI
C

EGFR

p.L858R (c.2573T>G)

Chr7:55,191,822 · Exon 21 · missense

Pathogenic

COSMIC ID

COSV55497715

COSM6224

FREQUENCY IN TUMOR TYPE

~43.0%

1,842 / 4,280 samples

ClinVar

Pathogenic

5★ · 14 submitters

VAF (TUMOR)

38.2%

coverage 1,240×

GENE ROLE

Oncogene

Census Tier 1

CMC SCORE

0.98

driver confidence

■ T790M secondary mutation confers resistance to 1st/2nd gen EGFR-TKIs — osimertinib preferred

Treatment options

1

Osimertinib (Tagrisso)

EGFR L858R · NSCLC 1L/2L · FDA approved 2018

FDA approved

1

Erlotinib / Gefitinib

EGFR L858R · 1G TKI · resistance risk with T790M co-mutation

FDA approved

2A

Amivantamab + lazertinib

Post-osimertinib progression · NCT04080836

Clinical trial

TP53

p.R248W (c.742C>T)
Chr17:7,674,220 · Exon 7 · missense

Pathogenic

COSMIC ID
COSV52765807

FREQUENCY IN TUMOR TYPE
~3.1%
133 / 4,280 samples

ClinVar
Pathogenic
5★ · 22 submitters

VAF (TUMOR)
61.8%
coverage 980×

GENE ROLE
Tumor suppressor
Census Tier 1

CMC SCORE
0.91
driver confidence

Treatment options

3

APR-246 (Eprentapopt)
TP53 gain-of-function mutations · phase II/III trials

Investigation

2B

Immunotherapy consideration
TP53 co-mutation may increase IO response in EGFR+ NSCLC — emerging data

Data emerg

STK11

p.Q369* (c.1105C>T)
Chr19:1,221,464 · Exon 8 · nonsense

Pathogenic

COSMIC ID
COSV56781203

FREQUENCY IN TUMOR TYPE
~8.4%
360 / 4,280 samples

ClinVar
Pathogenic
4★ · 9 submitters

VAF (TUMOR)
44.1%
coverage 860×

GENE ROLE
Tumor suppressor
Census Tier 1

CMC SCORE
0.87
driver confidence

■ STK11 loss associated with reduced PD-L1 immunotherapy response in NSCLC

Treatment options

2A

Pembrolizumab — reduced benefit
STK11 loss associated with IO resistance; consider combination strategies

Caut

2B

SHP2 inhibitor combination
Pre-clinical + early clinical data in STK11-null KRAS-co-mutant NSCLC

Investigation

KEAP1

p.R272C (c.814C>T)
Chr19:10,486,351 · Exon 4 · missense

VUS

COSMIC ID
COSV99102841

FREQUENCY IN TUMOR TYPE
~1.2%
51 / 4,280 samples

ClinVar
VUS
2★ · 4 submitters

VAF (TUMOR)
22.3%
coverage 1,100×

GENE ROLE
Tumor suppressor
Census Tier 1

CMC SCORE
0.44
driver confidence

Treatment options

3

No approved therapy
KEAP1/NRF2 pathway is an emerging target. May affect oxidative stress and chemotherapy sensitivity.

Emerg

Mutational signatures

SBS4 — Tobacco smoking

Associated with tobacco carcinogens; C>A transversions



SBS1 — Age-related

Spontaneous deamination of 5-methylcytosine



SBS2 — APOBEC activity

Cytidine deaminase activity; C>T at TCA/TCT context



SBS13 — APOBEC activity

Paired with SBS2; C>G at TCA/TCT context



Dominant tobacco signature (SBS4) consistent with smoking history. APOBEC signatures (SBS2/13) suggest ongoing mutational processes — may influence immunotherapy response.

Evidence level guide (OncoKB)

- 1** FDA-recognized biomarker in this tumor type
- 2A** Standard care biomarker in another tumor type
- 3** Clinical evidence in this tumor type (early)
- R** Known resistance to therapy

Data sources
Somatic Variants: COSMIC v99 (commercial license) · **Variant Classification:** ClinVar 2026-05 (NCBI) · **Actionability:** OncoKB v4.x + CIViC (accessed 2026-05-28) · **Population Frequency:** gnomAD v4.1

Disclaimer: This report is provided for informational and research purposes only. COSMIC data is used under commercial license and is not approved for clinical decision-making as a standalone resource. All treatment decisions must be made by a qualified oncologist in conjunction with full clinical context. Variant interpretations reflect current evidence as of report date and may change as new data emerge. ClinVar classifications represent community consensus and may differ from institutional interpretations. This sample report contains illustrative data only — no real patient data is included.